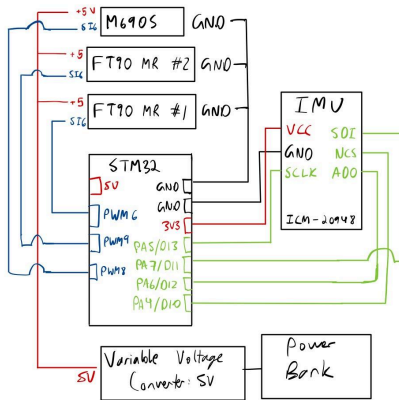
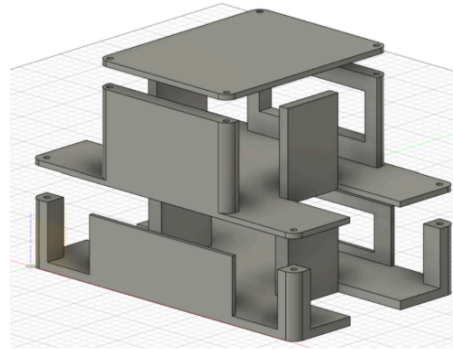


Dynamic Stability Cart for Rope-Bridge Traversal

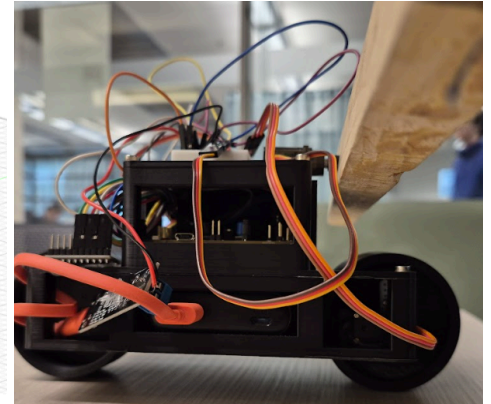
[Final Report](#) | [Demo](#) | [Cart Overview](#)



System Diagram (STM32 & connections)



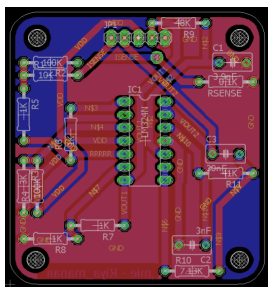
Deconstructed view of Cart



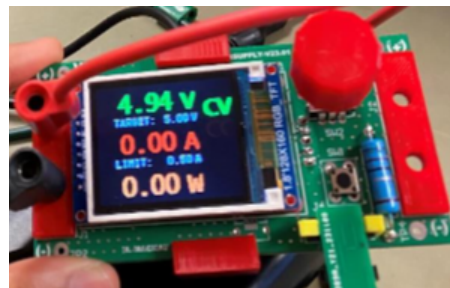
Side View of Cart with Balancing Beam

- Engineered a two-wheeled autonomous cart to traverse a custom-built rope-suspended bridge, integrating an **STM32** NUCLEO-G070RB with a 9-DoF ICM-20948 IMU over **SPI** and a servo-actuated balancing rod; developed embedded C++ firmware implementing PI control for real-time pitch stabilization, achieving $<3^\circ$ steady-state error and consistent successful bridge crossings on a 1 m single-axis rail.
- Reduced control-loop latency by $>95\%$ ($5s \rightarrow <100\text{ ms}$) by offloading quaternion-based sensor fusion to the IMU's onboard DMP, eliminating tip-over events caused by delayed pitch correction.
- Developed two **Fusion 360** design iterations of 3D-printed chassis, transitioning from a high-wall enclosed structure to an open multi-level layout that reduced mass while maintaining structural integrity; enforced chassis symmetry to center the CoM on the cart's longitudinal axis, simplifying balancing problem and improving disturbance rejection on 1-axis rail.
- Validated system stability through iterative hardware testing across progressively harder rail configurations (dual-anchor to single-anchor string attachment), diagnosing Kd-induced oscillations that caused tip-overs and simplifying to a tuned PI controller ($K_p=1.5$, $K_i=0.02$) to achieve repeatable autonomous bridge traversal.

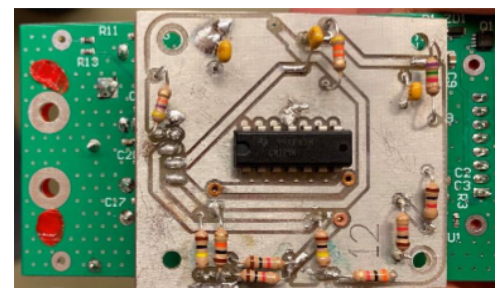
Current Sensing for Adjustable DC Power Supply



PCB Layout in EAGLE



Soldered PCB with LCD Display Active



PCB Integrated with STM32-based Power Supply

- Designed a low-side current sense amplifier in **PSpice** with non-inverting summing topology and op-amp buffer offset stage, mapping 0–3.5 A load current to a 0.25–3.3 V output range for STM32 ADC input; developed a 3rd-order 5 kHz low-pass filter with 35 dB attenuation to reject switching-regulator noise & prevent ADC aliasing.
- Translated analog schematic to a custom 2-layer PCB in **EAGLE** (layout, Gerbers, BOM), applying clean analog layout practices including ground pour, short high-current traces, and component keepout regions; soldered and assembled the board, validating DC operating point against PSpice predictions.
- Integrated current sense PCB with an STM32G051 microcontroller, building calibration firmware to map ADC readings to load current ($R^2=0.999$) and implementing real-time voltage/current control with rotary encoder and LCD interface.
- Validated system performance using oscilloscope measurements of filtered and unfiltered outputs under step-load conditions, confirming -35 dB filter attenuation at 5 kHz and $\leq 250\text{ mV}$ output noise; diagnosed ground-related ripple near cutoff and documented simulated vs. measured discrepancies with root-cause analysis.